

Laporan Dokumentasi Deployment

Dokumen ini menjelaskan arsitektur dan konfigurasi *deployment* aplikasi "Wondr Desktop" ke lingkungan produksi, yang menggunakan Google Kubernetes Engine (GKE) dan layanan Google Cloud lainnya.

1. Arsitektur Aplikasi

Aplikasi "Wondr Desktop" terdiri dari empat komponen utama yang terpisah dalam direktori *root* proyek:

- **backend-secure-onboarding-system**: Layanan *backend* utama.
- **frontend-secure-onboarding-system**: Aplikasi *frontend* berbasis web.
- **ops-secure-onboarding-system**: Direktori untuk konfigurasi operasional.
- **verificator-secure-onboarding-system**: Layanan verifikasi pihak ketiga.

```
.
├── backend-secure-onboarding-system
├── frontend-secure-onboarding-system
├── ops-secure-onboarding-system
└── verificator-secure-onboarding-system
```

1.1 Struktur Backend & Frontend

Struktur folder **backend** menunjukkan arsitektur Spring Boot dengan fokus pada layanan otentikasi, registrasi, dan verifikasi.

```
.
├── Dockerfile
├── alert-rules.yml
├── alertmanager.yml
├── mvnw
├── pom.xml
├── prometheus.yml
└── src
    ├── main
    │   ├── java
    │   │   ├── com
    │   │   │   ├── reg
    │   │   │   │   ├── regis
    │   │   │   │   │   ├── RegistrationAbsoluteApplication.java
    │   │   │   │   │   ├── client
    │   │   │   │   │   │   ├── DukcapilWebClient.java
    │   │   │   │   │   ├── config
    │   │   │   │   │   │   ├── DukcapilClientConfig.java
    │   │   │   │   │   │   ├── OtpConfig.java
    │   │   │   │   │   │   ├── RateLimitConfig.java
    │   │   │   │   │   │   ├── SecurityConfig.java
    │   │   │   │   │   │   └── SwaggerConfig.java
```

```

├── WebConfig.java
├── controller
│   ├── LoginController.java
│   ├── OtpController.java
│   ├── RegistrationController.java
│   └── VerificationController.java
├── dto
│   ├── request
│   │   ├── DukcapilRequestDto.java
│   │   ├── EmailVerificationRequest.java
│   │   ├── NikVerificationRequest.java
│   │   ├── PhoneVerificationRequest.java
│   │   └── RegistrationRequest.java
│   └── response
│       ├── DukcapilResponseDto.java
│       ├── RegistrationResponse.java
│       └── VerificationResponse.java
├── model
│   ├── Alamat.java
│   ├── Customer.java
│   └── Wali.java
├── repository
│   └── CustomerRepository.java
├── security
│   ├── JwtAuthFilter.java
│   ├── JwtUtil.java
│   └── SecurityUtil.java
├── service
│   ├── CustomerUserDetailsService.java
│   ├── DukcapilClientService.java
│   ├── LoginAttemptService.java
│   ├── RegistrationService.java
│   └── VerificationService.java
├── resources
│   └── application.properties
├── test
│   ├── java
│   │   ├── com
│   │   │   ├── reg
│   │   │   │   ├── regis
│   │   │   │   │   ├── TestSummary.md
│   │   │   │   │   ├── controller
│   │   │   │   │   │   ├── LoginControllerTest.java
│   │   │   │   │   │   ├── RegistrationControllerTest.java
│   │   │   │   │   │   └── VerificationControllerTest.java
│   │   │   │   │   ├── dto
│   │   │   │   │   │   ├── request
│   │   │   │   │   │   │   ├── DukcapilRequestDtoTest.java
│   │   │   │   │   │   │   ├── EmailVerificationRequestTest.java
│   │   │   │   │   │   │   ├── NikVerificationRequestTest.java
│   │   │   │   │   │   │   ├── PhoneVerificationRequestTest.java
│   │   │   │   │   │   │   └── RegistrationRequestTest.java
│   │   │   │   │   │   └── response
│   │   │   │   │   │       └── DukcapilResponseDtoTest.java

```

```

├── RegistrationResponseTest.java
├── VerificationResponseTest.java
├── model
│   ├── AlamatTest.java
│   ├── CustomerTest.java
│   └── WaliTest.java
├── repository
│   └── CustomerRepositoryTest.java
├── security
│   ├── JwtAuthFilterTest.java
│   ├── JwtUtilTest.java
│   └── SecurityUtilTest.java
├── service
│   ├── CustomerUserDetailsServiceTest.java
│   ├── DukcapilClientServiceTest.java
│   ├── LoginAttemptServiceTest.java
│   ├── RegistrationServiceTest.java
│   └── VerificationServiceTest.java
├── resources
│   └── application-test.properties

```

Struktur folder **frontend** menggunakan React dengan Vite. Konfigurasi Nginx disertakan untuk melayani aplikasi statis.

```

.
├── Dockerfile
├── eslint.config.js
├── index.html
├── nginx
│   └── default.conf
├── package-lock.json
├── package.json
├── public
│   ├── index.html
│   ├── vite.svg
│   └── wondr-logo.png
├── src
│   ├── App.jsx
│   ├── Dashboard
│   │   ├── Dashboard.jsx
│   │   └── Dashboard.module.css
│   ├── Login
│   │   ├── LoginForm.jsx
│   │   └── LoginForm.module.css
│   ├── assets
│   │   ├── Dashboard.png
│   │   ├── ...
│   │   └── wondr.png
│   ├── components
│   │   ├── PasswordForm.jsx
│   │   └── ...

```

```
├── undang.pdf
├── context
│   ├── RegisterContext.jsx
│   └── formContext.jsx
├── firebase.js
├── index.css
├── main.jsx
├── pages
│   ├── AccountConfirmation.jsx
│   ├── AddressInputPage.jsx
│   ├── CreatePassword.css
│   ├── CreatePasswordPage.jsx
│   ├── EKTPVerification.css
│   ├── EKTPVerificationPage.jsx
│   ├── EmailInput.css
│   ├── Home.css
│   ├── Home.jsx
│   ├── JenisKartuPage.css
│   ├── JenisKartuPage.jsx
│   ├── JenisTabunganPage.css
│   ├── JenisTabunganPage.jsx
│   ├── JumlahGaji.jsx
│   ├── MotherName.css
│   ├── MotherNamePage.jsx
│   ├── OccupationPage.jsx
│   ├── PenghasilanPage.jsx
│   ├── PersonalDataForm.css
│   ├── PersonalDataForm.jsx
│   ├── PhoneInputPage.css
│   ├── PhoneInputPage.jsx
│   ├── WaliIdentity.css
│   ├── WondrLanding.css
│   ├── WondrLanding.jsx
│   ├── address.css
│   ├── emailInputPage.jsx
│   ├── nameInput.css
│   ├── nameInputPage.jsx
│   ├── phoneOtpInputPage.css
│   ├── phoneOtpInputPage.jsx
│   ├── termsCondition.jsx
│   ├── termsConditions.module.css
│   ├── tujuanPembukaanRekening.jsx
│   ├── undang.jsx
│   ├── undang.module.css
│   ├── waliFormPages.css
│   ├── waliIdentityPage.jsx
│   └── waliPage.jsx
├── utils
│   ├── sanitize.js
│   └── validation.js
├── vite.config.js
└── wondr-logo.png
```

2. Infrastruktur Deployment

Aplikasi ini di-deploy menggunakan kombinasi GKE, Cloud Run, dan Cloud SQL untuk memastikan skalabilitas, efisiensi, dan manajemen yang mudah.

2.1 Google Kubernetes Engine (GKE)

- **Nama Cluster:** `gke-secure-onboarding-system`
- **Spesifikasi:**
 - **Tipe:** `Autopilot` (secara otomatis mengelola node dan pods).
 - **Versi:** `1.28.6-gke.1130000`.
 - **Lokasi:** `asia-southeast1-a` (Jakarta).
- **Deskripsi:** Cluster GKE digunakan untuk menampung `deployment backend-secure-onboarding-system` dan `frontend-secure-onboarding-system` serta database PostgreSQL.

2.2 Komunikasi Antar-Komponen dalam Cluster

- **Namespace:** *Deployment* dipisahkan ke dalam *namespace* `frontend` dan `backend`. Database PostgreSQL berada di *namespace* `backend`.
- **Network Policy:** Komunikasi antar *namespace* diatur dengan `NetworkPolicy`. Berikut adalah kebijakan yang memungkinkan *traffic* dari *namespace* `frontend` ke *service backend* dengan port `8080`.

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-frontend-to-backend
  namespace: backend
spec:
  podSelector:
    matchLabels:
      app: backend
  policyTypes:
    - Ingress
  ingress:
    - from:
        - namespaceSelector:
            matchLabels:
              name: frontend-app
          podSelector:
            matchLabels:
              app: frontend-app
      ports:
        - protocol: TCP
          port: 8080
```

2.3 Deployment & Service Kubernetes

a. Backend Deployment

- **Nama:** `backend-secure-onboarding-system-deployment`.
- **Replicas:** 2.
- **Image:** `asia.gcr.io/primeval-rune-467212-t9/wondr-desktop-be:1.0`.
- **Service:** `backend-service` (Tipe: `ClusterIP`) yang mengekspos port `8080` untuk komunikasi internal.

b. Frontend Deployment

- **Nama:** `frontend-secure-onboarding-system-deployment`.
- **Replicas:** 2.
- **Image:** `asia.gcr.io/primeval-rune-467212-t9/wondr-desktop-fe:1.0`.
- **Service:** `frontend-service` (Tipe: `NodePort`) yang mengekspos port `80` untuk diakses melalui Ingress.

c. Database Deployment

- **Nama:** `postgresql-deployment`.
- **Image:** `postgres:13.3-alpine`.
- **Service:** `postgresql-service` (Tipe: `ClusterIP`) untuk akses internal dari *backend*.

2.4 Ingress

- **Nama:** `secure-onboarding-ingress`.
- **Namespace:** `frontend`.
- **Konfigurasi:** Menggunakan Ingress GCE dengan IP statis global bernama `secure-onboarding-ip`.
- **Tujuan:** Meneruskan semua *traffic* dari domain `wondrdesktop.my.id` ke `frontend-service` pada port `80`.

2.5 Layanan Eksternal

Aplikasi ini menggunakan layanan eksternal untuk fungsionalitas tertentu:

- **Verifikasi OTP SMS:** Menggunakan layanan dari **Firebase Auth**.
- **Verifikator Dukcapil:** Layanan ini di-*deploy* sebagai **Cloud Run Service** dengan **Cloud SQL** sebagai databasenya. Hal ini dilakukan untuk memisahkan layanan yang spesifik dari *cluster* utama, memastikan skalabilitas yang lebih baik untuk layanan yang sifatnya terisolasi.
 - **Nama Layanan:** `verifikator-secure-onboarding-system`
 - **Endpoint:** `https://verifikator-secure-onboarding-system-441501015598.asia-southeast1.run.app`

Informasi Detail

informasi detail dari gke-cluster :

✔ gke-secure-onboarding-system



Automatic upgrades and other Google maintenance tasks may run at any time, increasing transient disruptions to your workloads. Set window.

[View details](#)

- Details
- Nodes
- Storage
- Observability
- Logs
- App Errors (10)

Cluster basics

Name	gke-secure-onboarding-system	🔒
Tier ?	Standard	✎
Mode ?	Standard	🔒
Location type	Zonal	🔒
Control plane zone	asia-southeast1-a	🔒
Default node zones ?	asia-southeast1-a	✎
Total size	4	①
Release channel ?	Regular channel	✎
Version ?	1.33.2-gke.1111000	✎ Upgrade available
Current COS version	cos-121-18867-90-62	①
End of standard support ?	Aug 3, 2026	①
End of extended support ?	Jun 3, 2027	①
Rollout sequence ?	To use rollout sequencing, register your cluster to a fleet	🔒

informasi backend-deployment:

✔ backend-deployment

Overview	Details	Observability	Revision history	Events
Cluster	gke-secure-onboarding-system			
Namespace	backend			
Created	Aug 3, 2025, 9:44:45 AM			
Labels	No labels set			
Annotations	deployment.kubernetes.io/revision: 3 ▼ Show all annotations			
Replicas	1 updated, 1 ready, 1 available, 0 unavailable			
Label selector	app = backend			
Update strategy ?	Rolling update, Max unavailable: 25%, Max surge: 25%			
Min time ready before available	0 s			
Progress deadline	600 s			
Revision history limit	10			

infromasi frontend-deployment:

✔ frontend-deployment

Overview	Details	Observability	Revision history	Events	Logs	App Errors (0)	YAML
Cluster	gke-secure-onboarding-system						
Namespace	frontend						
Created	Aug 3, 2025, 12:12:17 PM						
Labels	app: frontend-app						
Annotations	deployment.kubernetes.io/revision: 2 ▼ Show all annotations						
Replicas	1 updated, 1 ready, 1 available, 0 unavailable						
Label selector	app = frontend-app						
Update strategy ⓘ	Rolling update, Max unavailable: 25%, Max surge: 25%						
Min time ready before available	0 s						
Progress deadline	600 s						
Revision history limit	10						

informasi postgresql-deployment:

✔ postgresql

DetailsEventsLogsYAML

CPU ⓘ

1.5

1

0.5

0

UTC+710:40 AM10:50 AM11:00 AM11:10 AM11:20 AM

Cluster	gke-secure-onboarding-system
Namespace	backend
Created	Jul 30, 2025, 1:57:30 PM
Labels	app: postgresql
Annotations	▼ Show annotations
Logs ⓘ	Container logs , Audit logs
Label selector	app = postgresql
Replicas	1
Observed generations	1
Service	postgresql-service
Persistent volume claims specification	postgres-data

informasi **backend-service**:

✔ backend-service

Overview

Details

Events

Logs

App Errors (0)

YAML

Cluster	gke-secure-onboarding-system
Namespace	backend
Created	Aug 3, 2025, 9:44:45 AM
Labels	No labels set
Annotations	cloud.google.com/neg: {"ingress":true} ▼ Show all annotations
Label selector	app = backend
Pods	1 current / 1 desired
Type	ClusterIP

Cluster IP

Cluster IP	34.118.236.255
------------	----------------

informasi **frontend-service**:

✔ frontend-service

Overview

Details

Events

Logs

App Errors (0)

YAML

Cluster	gke-secure-onboarding-system
Namespace	frontend
Created	Jul 30, 2025, 4:13:27 PM
Labels	No labels set
Annotations	cloud.google.com/neg: {"ingress":true} cloud.google.com/neg-status: {"network_endpoint_groups":{"80":"k8s1-3963eb42-frontend-frontend-service-80-ed8857e4"},"zones":["asia-southeast1-a"]} ▼ Show all annotations
Label selector	app = frontend-app
Pods	1 current / 1 desired
Type	ClusterIP

Cluster IP

Cluster IP	34.118.230.170
------------	----------------

informasi **postgresql-service**:

✓ postgresql-service

Overview

Details

Events

Logs

App Errors (0)

YAML

Cluster

[gke-secure-onboarding-system](#)

Namespace

backend

Created

Jul 30, 2025, 1:57:50 PM

Labels

app: postgresql

Annotations

ccloud.google.com/neg: {"ingress":true}

✓ Show all annotations

Label selector

app = postgresql

Pods

1 current / 1 desired

Type

ClusterIP

Cluster IP

34.118.239.149

informasi **secure-onboarding-ingress**:

✓ secure-onboarding-ingress

Details

Telemetry

Logs

Events

YAML

Cluster

[gke-secure-onboarding-system](#)

Namespace

frontend

Created

Aug 3, 2025, 12:31:22 PM

Labels

No labels set

Annotations

ingress.kubernetes.io/backends: {"k8s1-3963eb42-frontend-frontend-service-80-ed8857e4":"HEALTHY", "k8s1-3963eb42-kube-system-default-http-backend-80-be03669c":"HEALTHY"}

ingress.kubernetes.io/forwarding-rule: k8s2-fr-ge1n6rz1-frontend-secure-onboarding-ingress-m6uyswvy

ingress.kubernetes.io/target-proxy: k8s2-tp-ge1n6rz1-frontend-secure-onboarding-ingress-m6uyswvy

ingress.kubernetes.io/url-map: k8s2-um-ge1n6rz1-frontend-secure-onboarding-ingress-m6uyswvy

kubernetes.io/ingress.class: gce

kubernetes.io/ingress.global-static-ip-name: secure-onboarding-ip

✓ Show all annotations

Type

Ingress

Load balancer

External HTTP(S) LB

IP address

[34.49.249.12](#)

Routes

Route	Service	Pod selector	Clusters	Pods
wondrdesktop.my.id/*	frontend-service	app = frontend-app	gke-secure-onboarding-system	1/1

Serving pods

Service	Name	Status	Restarts	Created on
frontend-service	frontend-deployment-69fb7bc8c7-6jcpq	✓ Running	0	Aug 4, 2025, 11:21:07 AM

Load balancing resources

Load balancer

[k8s2-um-ge1n6rz1-frontend-secure-onboarding-ingress-m6uyswvy](#)

Target proxies

[k8s2-tp-ge1n6rz1-frontend-secure-onboarding-ingress-m6uyswvy](#)

Forwarding rules

[k8s2-fr-ge1n6rz1-frontend-secure-onboarding-ingress-m6uyswvy](#)

Backend services

✓ [k8s1-3963eb42-frontend-frontend-service-80-ed8857e4](#)

✓ [k8s1-3963eb42-kube-system-default-http-backend-80-be03669c](#)

informasi service **verificator-secure-onboarding-system** :

verificator-secure-onboarding-system

Region: asia-southeast1

URL: <https://verificator-secure-onboarding-system-441501015598.asia-southeast1.run.app>

MetricsSLOsLogsRevisionsTriggersNetworkingSecurityYAML

RevisionsManage traffic

FilterFilter revisions

	Name	Traffic	D	Actions
☒	verificator-secure-onboarding-system-00019-njt	100% (to latest)	2	
☐	verificator-secure-onboarding-system-00018-cg9	0%	1	
☐	verificator-secure-onboarding-system-00017-mpf	0%	4	
☐	verificator-secure-onboarding-system-00016-fpz	0%	4	
☐	verificator-secure-onboarding-system-00012-47t	0%	4	
☐	verificator-secure-onboarding-system-00008-r2k	0%	6	

verificator-secure-onboarding-system-00019-njt

Deployed by bostangsteitb2020@gmail.com using Cloud Console

ContainersVolumesNetworkingSecurityYAML

General

Billing	Request-based
Startup CPU boost	Enabled
Concurrency	80
Request timeout	300 seconds
Execution environment	Default

Autoscaling

Revision max instances	100
------------------------	-----

Image

asia.gcr.io/primeval-rune-467212-t9/verificator-sec...

Port

8080

Build

(no build information available)

Source

(no source information available)

Command and args

(container endpoint)

CPU limit

1

Memory limit

512MiB

informasi wondr-desktop-otp:

Firebase

Project Overview

Project shortcuts

Authentication

What's new

AI LogicNEW

Product categories

Build

Run

Analytics

AI

Related development tools

Firebase Studio

wondr-desktop-otp

Project settings

GeneralCloud MessagingIntegrationsService accountsData privacyUsers and permissionsAlerts

Your project

Project name

wondr-desktop-otp

Project ID

wondr-desktop-otp

Project number

15939231912

Web API Key

my4j6_hPQR_7eFyo

Environment

This setting customizes your project for different stages of the app lifecycle

Environment type

Unspecified

11 / 11